WHEN BACKGROUND NOISE HAS MEANING

Environmental sound classification under semantic interference

Qihan Zhu · SID 550489593

Same target. Same relative level. Different interference.

2,000 ESC-50 clips · official five-fold evaluation · paired mixtures for every model

White noise

Acoustically nonspecific

DEMAND

Traffic · station · metro

Same category

Different fine class

Different category

Different coarse category

+10 dB

Weaker interference

+5 dB

Primary comparison

0 dB

Equal-power stress test

24,000 own-activity mixtures; 23,991 target-activity mixtures after three undefined pairs × three levels are excluded.

Literature and course foundations

Each method addresses a different part of the research question

ESC-50 baselines	Historical SVM / CNN: 39.6% / 64.5% accuracy Ours: 48.9% / 67.9%; different pipelines, not a controlled gain
Between-class learning	Mixture proportions become soft training targets Our semantic augmentation keeps one designated target label
CLAP / PAT	Paper: 88–91% with overlap exclusions; ours: 81.75% Different checkpoint/protocol; fixed ensemble is not full PAT
Course DSP / research gap	STFT, features, SNR/SIR and known distractor identity Measure where the decision moves under identical mixtures

Report Section 2 compares strengths, limitations and the reason for selecting each representation.

Three representations, one controlled experiment

Deployment behaviour is comparable; total pretraining history is not.

A **Handcrafted + SVM**

Training-fold standardisation

STFT → spectral descriptors
MFCCs + deltas → 92 features

B **Compact log-Mel CNN**

Three optimisation seeds

STFT → Mel filters → log power
36,258 trainable parameters

C **Frozen LAION-CLAP**

No training or fine-tuning

Audio / text embedding similarity
50 fixed class descriptions

Custom STFT checked against SciPy and actual course MATLAB code; maximum absolute discrepancy 2.93×10^{-14}.

Power matching depends on where power is measured

The original target-mask definition is retained as a full sensitivity analysis.



Own-activity power

Measure each source over its own active interval.

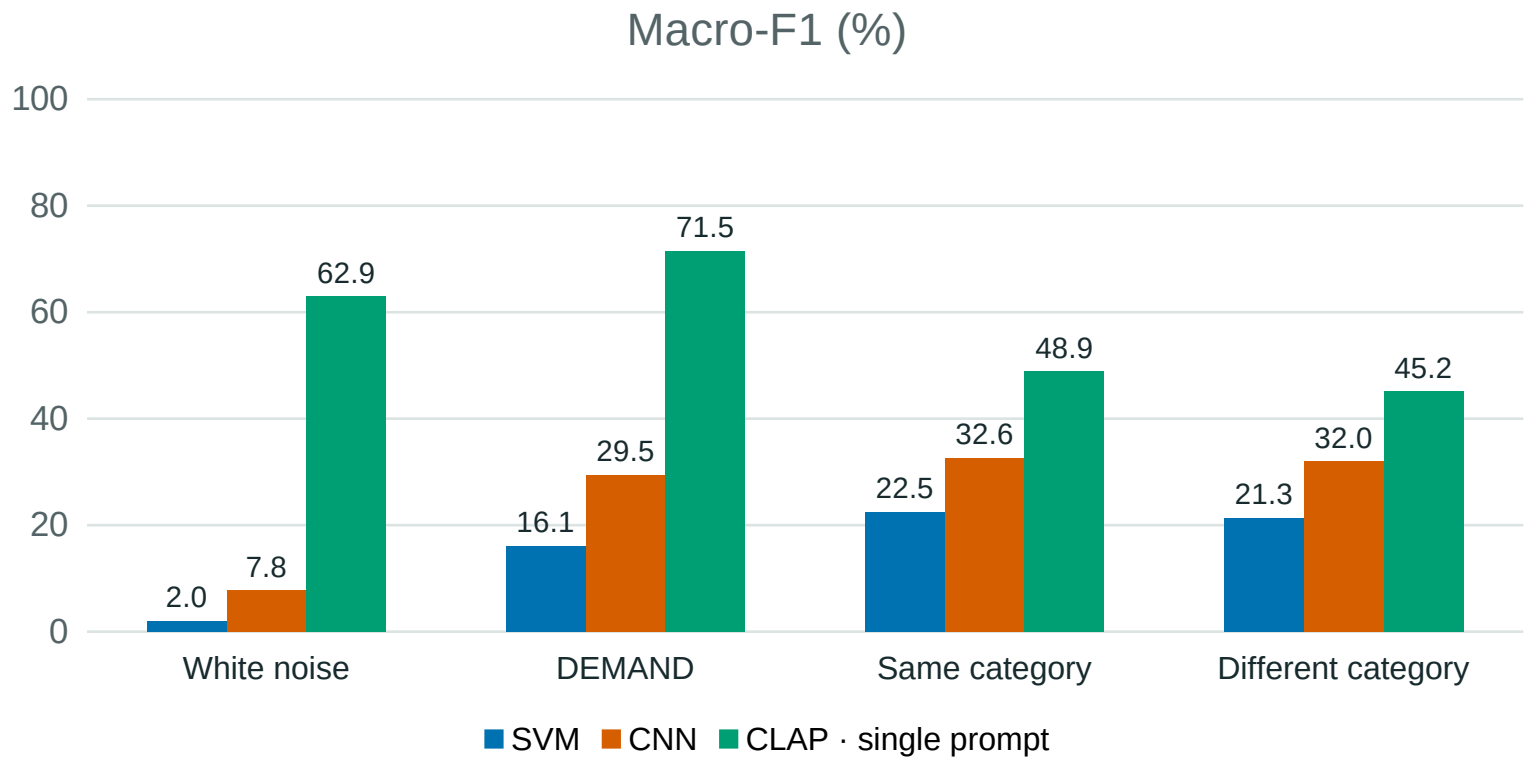
Target-activity power

Measure both sources over the target’s active interval.

Illustrative masks. Common peak attenuation prevents clipping; a gain-matched clean control isolates its effect.

Interference power alone does not predict failure

Own-activity protocol · +5 dB · five-fold mean macro-F1



Where do errors go?

Capture rate (%) · errors only

System	Same	Different
SVM	18.53	15.89
CNN	31.26	27.53
CLAP	66.95	60.30

Captures ÷ errors, pooled over folds.
Not captures ÷ 2,000 trials.

High-overlap CNN − DEMAND accuracy: −3.28 / −2.94 pp (n = 1,464 / 1,465); both Holm p = 0.0604. Original seed; same / different.

Does robustness training transfer to competing events?

Same augmentation probability and epoch cap · clean validation · three CNN seeds

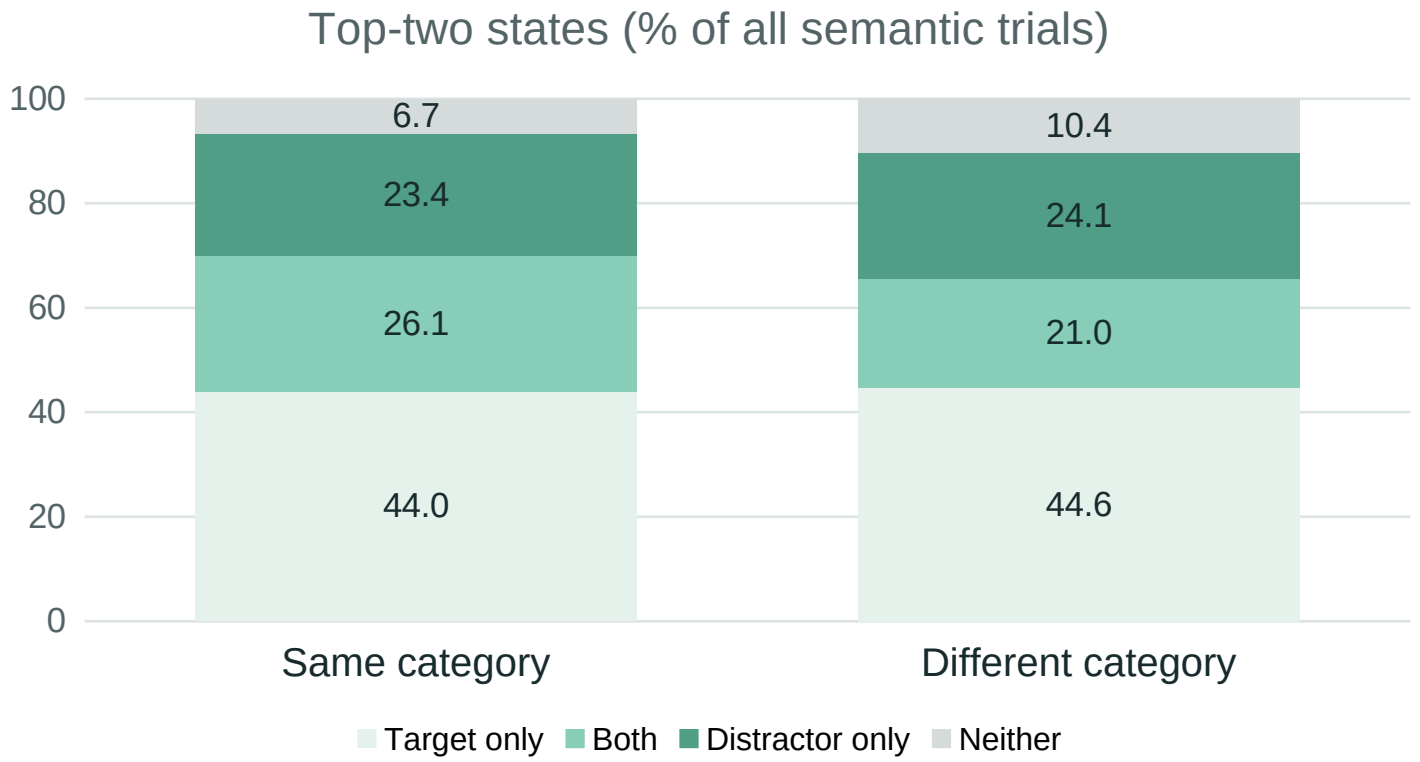
Model / training	Clean	White	DEMAND	Same	Different
SVM / clean	48.0 ± 4.1	2.0 ± 0.9	16.1 ± 2.0	22.5 ± 2.7	21.3 ± 2.6
SVM / WGN	46.0 ± 4.1	41.1 ± 5.4	23.7 ± 5.1	22.6 ± 3.4	22.2 ± 2.6
SVM / semantic	49.3 ± 2.4	1.9 ± 0.6	19.9 ± 2.6	26.8 ± 2.3	25.0 ± 0.9
CNN / clean	65.8 ± 0.7	7.5 ± 0.4	30.9 ± 1.4	33.0 ± 0.3	31.4 ± 0.5
CNN / WGN	64.6 ± 0.6	43.9 ± 2.7	33.8 ± 2.2	33.5 ± 1.0	32.1 ± 1.1
CNN / semantic	61.7 ± 2.4	9.9 ± 0.2	33.6 ± 1.6	36.1 ± 0.8	33.7 ± 0.9

Own activity · macro-F1 (%) · interference conditions at +5 dB

CNN: mean ± seed SD; SVM: mean ± fold SD. Direct CNN semantic-over-WGN accuracy gains do not pass the 156-test Holm correction.

A captured decision can retain the target at rank two

Single-template CLAP · +5 dB · own activity · 2,000 targets per category type



37.34% / 33.17%

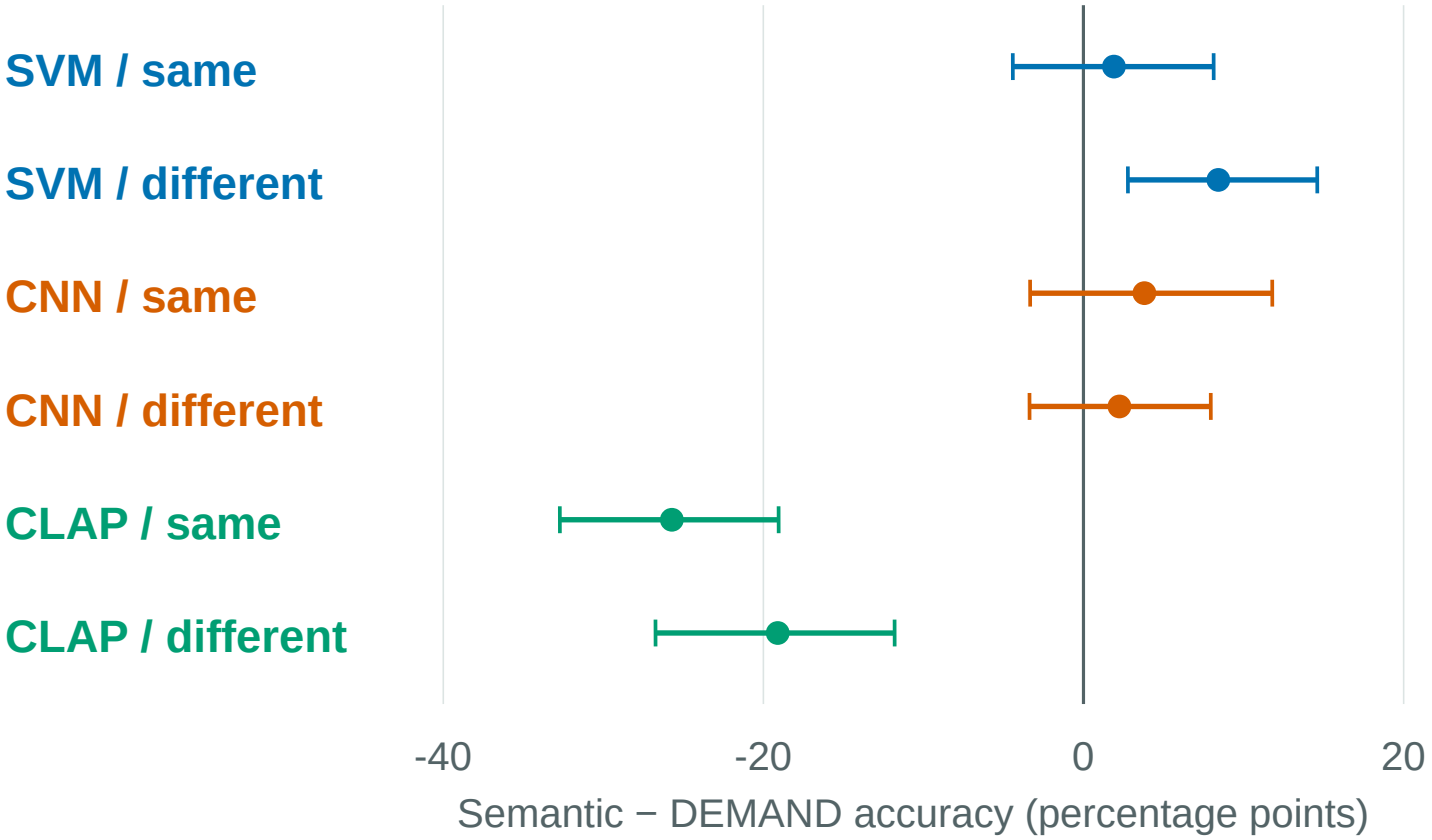
of original captures retain the target at rank two (same / different category)

Top-2 helps some cases. It does not prove that the model heard both sources.

Clean-control joint inclusion: 5.40% / 1.70%. Target-power results and k = 1, 3, 5 sensitivities are retained.

CLAP retains a gap in acoustically close subsets

Same-target comparison · +5 dB · own activity · primary calipers



CNN intervals include zero.
CLAP remains lower.

210 / 178 targets retained
(same / different category)
≈ 9–10.5% of the benchmark

Selected-population association; not proof of semantic causation.

Pointwise 95% target-source bootstrap intervals, 2,000 resamples. Both protocols: CLAP −16 to −26 pp. Subsets differ from high-overlap groups.

Real code execution and sound examples

Can opening + clock alarm · +5 dB SIR · full target-activity overlap

Gain-matched clean control

Play 5-second audio

Target + weaker clock alarm

Play 5-second audio

SVM

can opening
↓
chirping birds

Open the live code demonstration

CNN

can opening
↓
clock alarm

CLAP

can opening
↓
clock alarm

Recorded execution video

Discussion and conclusion

Representation, training and readout explain different parts of the evidence

01 Representation matters

CLAP displacement: correct 0.1905 / error 0.4087.
Own / +5 dB / same category; association, not cause.

02 Readout changes the answer

Some argmax captures retain the target
at rank two; many do not.

03 Acoustic controls narrow the claim

CLAP retains a gap in selected subsets;
this does not establish semantic causation.

Limits: three backgrounds · one pair per category type · pretraining overlap · 0-dB label ambiguity

Report, supplement and reproducible code

10-page report · 26-page supplement · working code · execution video

ESC-50 / Piczak	Official folds, dataset caveats and classic spectrogram CNN
BC learning / Tokozume et al.	Mixtures with proportional labels, ICLR 2018
LAION-CLAP / Wu et al.	ICASSP 2023; numerical context from revised arXiv v4
PAT / Seth et al.	Released prompt-list ablation; full PAT is not reproduced
Mix-CLAP / Kohno et al.	Noise-aware audio-language model context, ICASSP 2026
RSA-Bench / Zhang et al.	Structured-background speech/reasoning context, ACL Findings 2026

github.com/Charlie-Uni/elec5305-project-550489593

Audio: LG, Can Open.WAV (CC BY); hookhead, alarm clock.aif (CC0). ESC-50 compilation: CC BY-NC 3.0.

AI: OpenAI Codex / GPT-6 Astra; Anthropic Claude (version not recorded). Substantial code, analysis and drafting assistance; see report.